Muhammad Fahad Siddiqui

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COMPREHENSIVE MASTER CV / CAREER INVENTORY — October 2026

Not a job-specific application CV; selectively tailor and verify details before submission.

Professional Profile

Full-stack software engineer with approximately five years of cumulative software-development experience across SaaS, education technology, consumer applications, healthcare ERP, enterprise systems, and Web3. Currently a Technical Lead and hands-on Full-Stack Developer at Anue Education Solutions GmbH, combining software architecture, backend implementation, frontend integration, production operations, code review, hiring, and mentoring. Experienced in Java/Spring Boot, Node.js/TypeScript, React/Next.js, Angular, Python/Django, SQL databases, REST API design, cloud infrastructure, and CI/CD. Also building independent products that explore LLM integration, real-time conversational AI, Go services, and Rust-based event-driven systems. Comfortable owning the engineering lifecycle from requirement discovery and design through implementation, testing, deployment, and incident resolution.

Education

M.Sc. High Integrity Systems Apr 2024 – Jul 2026

Frankfurt University of Applied Sciences, Germany *Thesis pending*

- Relevant areas studied: IT Security; Data Mining; Learning from Data; Testing Methods; Advanced Real-Time Systems; Smart Sensor Network Systems; DBMS Implementation.
- Academic exposure to Python, Jupyter Notebook, machine-learning workflows, scikit-learn, data analysis, software testing, system dependability, and high-integrity software engineering.

B.S. Software Engineering Dec 2018 – Mar 2023

Sir Syed University of Engineering and Technology, Pakistan

- Foundations in software design, Java, object-oriented programming, algorithms, databases, web development, software engineering processes, and testing.

Technical Skills — Comprehensive Inventory

This is an inventory, not a claim of equal depth in every technology. Experience is differentiated in the evidence sections below. **Primary backend:** Java, Spring Boot, JavaScript, TypeScript, Node.js, NestJS, Python, Django, REST API design and documentation, authentication/authorization, integrations, application architecture.

Additional backend / systems: Go (Golang), Gin, Rust, PHP, Laravel, Express.js, Go concurrency (goroutines, channels, contexts, interfaces), Solana/Web3 integrations.

Frontend: React, Next.js, Angular, AngularJS, Vue.js, HTML5, CSS3, Tailwind CSS, shadcn/ui, responsive design, component systems, frontend/backend integration, React Native, Flutter, Dart.

Content / web: Astro, Sanity CMS, headless CMS, content schema modeling, landing pages, structured content integration, accessibility and performance considerations.

Databases / data: PostgreSQL, MySQL, MongoDB, SQL, Prisma, TypeORM, Liquibase, data cleaning/preparation, ETL concepts, data analysis, dataset integration, Jupyter Notebook.

AI / machine learning: OpenAI API, LLM APIs and integration, LangChainGo, RAG concepts and implementation, conversational AI, Vapi voice orchestration, scikit-learn, ML coursework, prompt design, AI-assisted coding (Claude Code, OpenAI Codex).

Data visualization / additional reported tools: D3.js, ECharts, Apache Spark; familiarity/exposure reported for TensorFlow, PyTorch and other ML workflows should be verified before external use.

Cloud / DevOps: AWS (EC2, S3), GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, Azure DevOps, CI/CD, Git, GitHub, staging and production deployments.

Messaging / observability: Kafka, RabbitMQ, Redis, MQTT (intermittent exposure), Sentry, Grafana, Prometheus, application logging, incident investigation.

Engineering practices: Agile delivery, system design, code reviews, mentoring, technical hiring/onboarding, API testing, Playwright, unit testing, debugging, release management, production support, stakeholder communication.

Professional Experience

Technical Lead & Full-Stack Developer **Mar 2025 – Present**

Anue Education Solutions GmbH Germany / remote collaboration **Product context:** Education-technology SaaS for course discovery, matching, applications, and related student journeys. Hands-on technical lead with engineering ownership across frontend, backend, infrastructure, and developer workflow.

- Lead technical decisions and delivery while remaining an individual contributor; review code, mentor and onboard junior engineers, participate in hiring, and coordinate technical priorities with business stakeholders.
- Develop Java Spring Boot and Python Django backend services, React/Next.js/TypeScript frontend features, and PostgreSQL persistence and queries; integrate user-facing functionality with REST services.
- Created and documented **300+ REST API endpoints** across product workflows, including course matching and favorites, applications, referrals, scoring, blog/content, shop, and related functionality.
- Implemented and maintained authentication-related flows, data-driven application features, and integrations with external services; supported payment integration with Mollie, including multiple payment methods.
- Worked with a catalog of approximately **26,000 courses**, including a smaller monetizable subset (previously reported at approximately 3,500); implemented and debugged backend data and course-discovery functionality.
- Developed application logic, database queries, data processing/preparation, and exploratory work using Python, SQL, and Jupyter Notebook where relevant.
- Managed AWS EC2/S3 infrastructure and Docker-based environments; maintained staging workflows, CI/CD with GitHub Actions, deployment troubleshooting, and database migrations with Liquibase.
- Investigated production and staging incidents across frontend, APIs, database migrations, and external services; used logs and Sentry, collaborated with users/stakeholders, and documented fixes and operational practices.
- Worked with container orchestration and monitoring/operations tools, including reported Kubernetes, Terraform, Grafana and Prometheus exposure; confirm exact environment and ownership before using these as quantified achievements.
- Used Claude Code and OpenAI Codex to accelerate implementation, debugging, refactoring, and testing while reviewing and validating generated code.
- Worked on web and mobile-facing workflows; reported professional Flutter/Dart exposure on a mobile application.

Full-Stack Engineer **Sep 2023 – Apr 2024**

MSE TechnologyPakistan **Product context:** Client-facing and enterprise applications, including work on the P&G Pampers Rewards ecosystem and a diaper-size tracking experience.

- Developed APIs and full-stack functionality with Node.js, NestJS, TypeScript, TypeORM, PostgreSQL, React/Next.js, and Angular, with responsibility for backend logic and frontend integration.
- Contributed to production backend services for the **P&G Pampers Rewards** product, including application workflows, data persistence, API development, and integrations.
- Worked with React Native and Flutter/Dart in mobile application contexts, including a Pampers Diaper Size Tracker project.
- Used **Sanity headless CMS**, including structured content schemas and integration of CMS-managed data into web applications.
- Participated in cloud, container, and CI/CD workflows; experience includes GCP, Azure, Docker, and reported Kubernetes exposure.
- Collaborated across implementation, debugging, data integration, and production delivery; also reported exposure to GCP-based machine-learning work.

Web3 / Blockchain Software Developer **Jun 2023 – Aug 2023**

Ashtar Global Solutions / Tumi LabsPakistan **Product context:** Solana-based blockchain game and associated web services, including the game **SolPatrol**.

- Developed **Go backend REST APIs** for authentication, wallet integration, transaction processing, and blockchain-related services; used Go interfaces, goroutines, channels, context handling, and unit tests.
- Worked with Rust and Solana blockchain integrations, including reported work involving Honeycomb Protocol.
- Built frontend functionality using Next.js, React, JavaScript/TypeScript, and Web3 integration patterns.
- Built and delivered the **SolPatrol production landing website using Astro**, connecting the game's presentation and web experience.
- Used Docker and reported Kubernetes exposure as part of the wider engineering environment.

Full-Stack Developer **Feb 2023 – May 2023**

Sidat Technologies *Pakistan*

- Developed web and mobile-facing application functionality using MERN technologies, Angular, TypeScript, and PHP/Laravel, including REST integrations and backend business logic.
- Built frontend components, forms, services, and database-backed workflows; participated in database optimization, debugging, and implementation across the SDLC.
- Used **Sanity CMS** professionally for headless content modeling, content schemas, and integration with application interfaces.
- Participated in Agile development, code review, CI/CD, and Azure DevOps-related workflows.

Junior Software Engineer / Full-Stack Developer **Aug 2022 – Jan 2023**
Binary Div *Pakistan*

- Worked on a Canadian medical ERP application using React, TypeScript/JavaScript, Node.js/NestJS, Prisma, and MySQL.
- Implemented UI features, REST APIs, database models and queries, and business workflows; supported testing and debugging of integrated application functionality.
- *Role-specific note:* Go backend work on an internal React Native mobile app has been discussed for a particular previous application, but is intentionally not treated as a general Binary Div claim without renewed confirmation.

Web Developer Intern **Jun 2021 – Aug 2021**
Core2Plus *Pakistan*

- Built responsive web interfaces and reusable UI elements using JavaScript, React, HTML, and CSS; participated in frontend development and team software-delivery practices.

Selected Professional Systems and Feature Ownership

Education SaaS — Anue

- **Backend platform:** Spring Boot services, PostgreSQL, documented REST APIs, authentication-related logic, validation, business workflows, integrations, and production debugging.
- **Course discovery and matching:** course data ingestion/integration, favorites, matching and scoring logic, similar courses, and student application workflows.
- **Growth and commerce:** referral program, shop features, payment-provider integration (Mollie), and supporting user-facing flows.
- **Engineering operations:** staging environments, CI/CD, deployment investigations, Liquibase migrations/checksum issues, database validation, S3/static asset handling, and external email template incidents.
- **Leadership:** technical planning, code review, mentoring, onboarding, hiring, architecture discussions, cross-functional coordination, and client-facing issue resolution.

Consumer / Enterprise / Healthcare Applications

- **Pampers Rewards:** backend APIs, business workflows, data persistence, and integration work at MSE Technology.
- **Pampers Diaper Size Tracker:** reported Flutter/Dart mobile development at MSE Technology.
- **Canadian medical ERP:** UI, API, and relational database workflows at Binary Div.
- **SolPatrol:** production Astro game landing page and Solana-oriented backend/frontend development at Ashtar.

Independent Engineering Projects

Mentora AI — Real-Time Conversational Tutoring SaaS

Stack: Next.js, TypeScript, Clerk, Supabase, Vapi, Sentry; AI voice services and subscription-aware product flows.

- Built a full-stack learning product where users create configurable AI tutors by subject, topic, voice, speaking style, and session duration.
- Integrated real-time speech/conversation experiences, authentication and user profiles, persistent session history, bookmarks, tutor discovery, and subscription-aware access patterns.
- Focus areas: low-latency conversational interfaces, SaaS architecture, external AI-service orchestration, and observability.

Plotlyst — AI-Powered Movie and TV Discovery *(in development)*

Stack: React, Go/Gin, MongoDB, LangChainGo, OpenAI API.

- Building a movie/TV discovery platform with search, structured metadata, recommendations, watchlists, curated rankings, actor information, and streaming-related discovery.
- Exploring natural-language movie discovery and LLM-backed recommendations, with a Go REST API and document database.
- Designed a mock-JSON data strategy to support engineering development when third-party movie-data APIs are unavailable or limited.

PitchPulse — Live Football Intelligence *(in development)*

Stack: Vue.js, Rust, Kafka, Redis, RabbitMQ; simulated match/event data.

- Designing a real-time football analytics interface with moving players and ball on a pitch, tactical event interpretation, and voice-based questions about match events.
- Exploring streaming/event-driven architecture, caching, message queues, simulation, and AI explanations of tactical sequences.
- Using simulated match coordinates and event data where live tracking-provider access is restricted.

SolPatrol — Production Game Landing Website

Stack: Astro, JavaScript/TypeScript, frontend web technologies.
Designed and delivered the landing website for a Solana-based game during professional work at Ashtar Global Solutions. *Production URL not yet recorded in the master inventory.*

Data, AI and Academic Experience

- **Jupyter / data preparation:** university coursework and reported professional usage at Anue; Python, SQL, exploratory data analysis, structured datasets, and data transformation.
- **Machine learning:** scikit-learn and ML coursework in data mining/learning-from-data contexts; GCP-related ML exposure at MSE. Specific production model-training responsibilities are not yet documented.
- **LLM integration:** OpenAI APIs, LangChainGo and AI-powered discovery; Vapi-powered conversational tutoring; practical AI-assisted software-development workflows.
- **Visualization:** D3.js and ECharts reported as hands-on tools; concrete employer/project examples and performance metrics should be added when confirmed.
- **Data platform tools:** Apache Spark reported as used; scope, environment, and project evidence still to be documented.
- **Model frameworks:** TensorFlow and PyTorch were mentioned in job-target discussions, but professional depth and specific implementations have not been established; do not automatically claim production proficiency.

Leadership, Architecture and Engineering Practices

- Hands-on technical leadership, architecture ownership, code reviews, mentorship, junior engineer hiring and onboarding, and coordination with non-technical stakeholders.
- API architecture, frontend/backend integration, authentication and authorization flows, database schema changes, migration troubleshooting, integration testing, and deployment workflows.
- Production ownership: investigate bugs and incidents using application logs and Sentry, trace issues across services, address deployment or database problems, and communicate resolutions.
- AI-assisted development with Claude Code and OpenAI Codex: initial implementation, code navigation, refactoring, tests, debugging, and architectural exploration; validate output through review and testing.
- Remote and distributed collaboration across teams, projects, and time zones.

Languages and Communication

English: IELTS Academic 7.0 (test taken Aug 2023; self-reported professional fluency).
German: B1 (self-reported, Oct 2026).
Urdu: Native.

Additional Context for Future Tailoring

- **Target positioning:** mid-level or senior full-stack/backend/software engineer depending on role; currently performs Technical Lead responsibilities. For Dataiku specifically, interested in a mid-level full-stack role.
- **Location:** Frankfurt am Main, Germany; remote work experience. For Germany applications use German contact information. A separate Pakistan-track CV may use Karachi, Pakistan and a Pakistan contact number when relocation is relevant.
- **Strongest narratives:** Java/Spring Boot and TypeScript full-stack SaaS delivery; API/backend ownership; AI integration; technical leadership; cloud/DevOps; Go and Rust/Web3 experience; Angular and React frontend work.
- **Evidence still worth collecting:** exact dates for Core2Plus and early roles, direct links to live projects, verified performance/latency/cost improvements, team sizes, customer usage, exact AI/ML models and outcomes, and depth of Spark/D3.js/ECharts/Kubernetes/Terraform usage.